Better Data Beats Big Data: A Case of Education

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Say "Big Data"
One More Time

MUCH ADO ABOUT "BIG DATA"

- "Big Data" is one of major ~~buzz words~~ cross-discipline fields of study
  - A lot of people heard of it
  - Emergence of a new field of Data Science

- Data sets approach population scale
  - Data is not cherry-picked
  - Statistical models fit to the big data are arguably more powerful/generalizable

- Not all data is created equal
  - A fraction of the [educational] data [collected by a tutor] ***can*** effectively represent the full data set

PROBLEM AT HAND: TAKING A TUTOR TO A NEW SCHOOL

- When
  - A new school adopts Cognitive Tutor
  - A school renews Cognitive Tutor subscription for another year
- Questions
  - Can we, in principle, tune CT to better serve its students?
    - Cognitive Tutor is driven by a cognitive skill model of the domain, better modeling means better learning experience
  - What information about the school or its students would help?
- Big Questions
  - Are there distinct *types* of schools (and students) in terms of student modeling parameters?
  - Can we *effectively** capture these differences?

* We will define what effectively means later

BEST WAY TO SLICE U.S. K-12 EDU SYSTEM

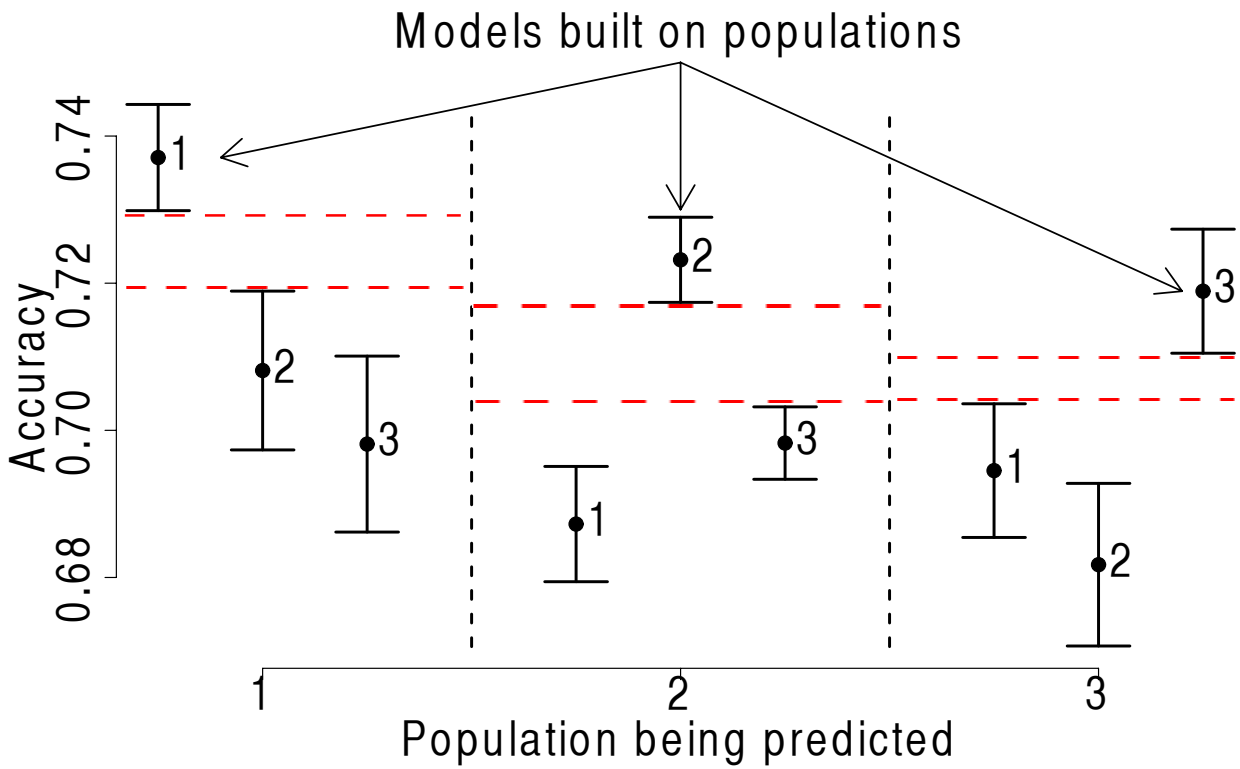
- National Center for Educational Statistics [http://nces.ed.gov]
  - School locale (rural, suburban, & urban)
  - Enrollment (school size)
  - Student-to-teacher ratio (proxy of inverse of teacher attention potential)
  - Number of students eligible for free or reduced price lunch (proxy of SES)
- Carnegie Learning Data
  - Student usage, usage variability, school-level coverage
- Logistic regression model
  - $\text{response}_{ij} = \text{student}_i + \text{problem}_j + \Sigma_k \text{skill_intercept}_k + \text{skill_slope}_k * \text{tries}_{ik}$
  - student intercept (student ability)
  - skill intercept (skill complexity)
  - skill slope (skill's *speed of learning*)

[BIG] DATA

- Carnegie Learning Cognitive Tutor data for 2010
  - ~144 000 registered students
  - 899 schools (approx. 20% of all schools)
  - ~470 000 000 records
- Cleanup and merging with NCES school data
  - Only problem-solving data
  - No practice problems
  - Multiple school accounts merged
  - Students completing 1 unit or less – removed
- Final dataset
  - ~230 schools
  - ~55 000 students
  - ~67 000 000 problem-solving records

DEFINING AN EFFECTIVE* SCHOOL GROUPING

- 3 ***hypothetical*** groups of schools
- Validate models across groups
  - From each group randomly draw non-overlapping 20 test and training set pairs
  - Train and test models within and between groups
  - Graph results



- Effective grouping is:
  - Model built on the training data of a particular group data predicts test data from the same group better than models build on the data from other groups
  - This is true for all group models

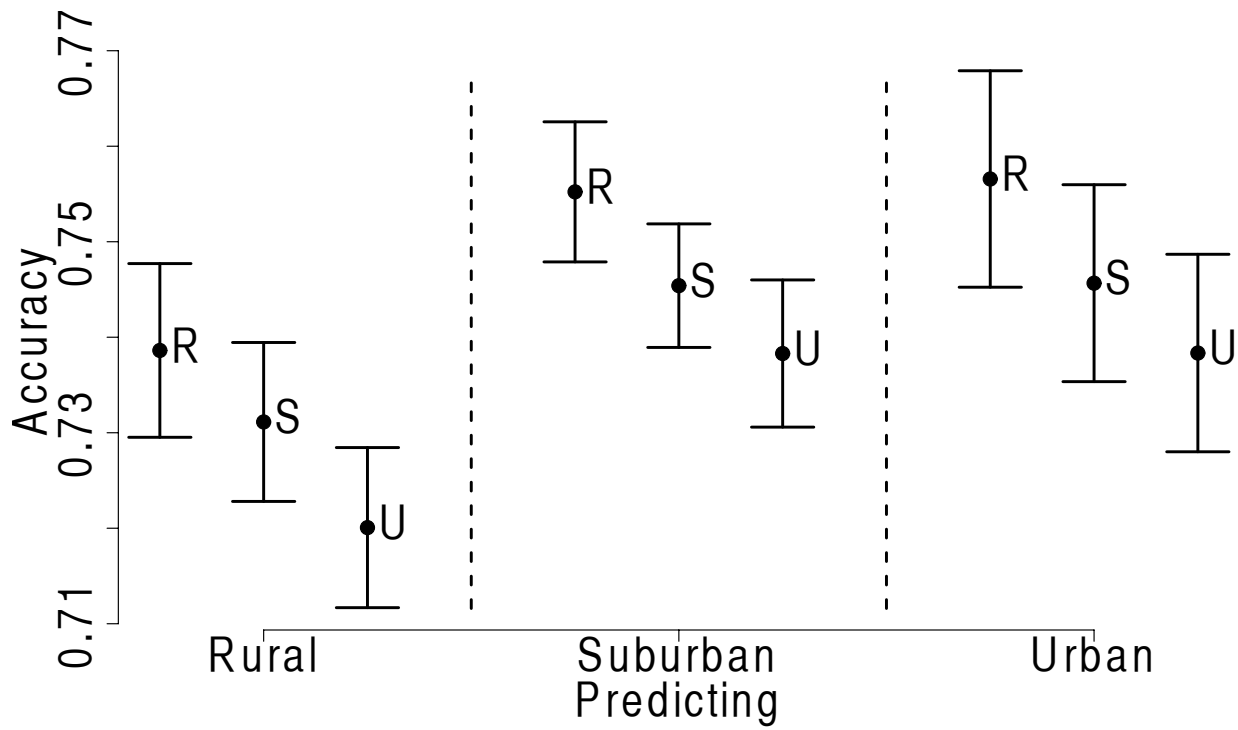
GROUPING FACTORS

Factor	Factor group
School locale*	School metadata, known a priori (NCES)
%Free & Reduced Lunch	
Student-teacher ratio	
Enrollment	
Avg. student-units attempted	Student usage data (Carnegie Learning)
SE student-units attempted	
School coverage group*	
Avg. student intercept	Logistic regression model values
Avg. skill intercept	
Avg. skill slope	

- All actor values are per-school
- All factors except two (*) are continuous
- Continuous factors are binned into 3 groups
  - No. students approx. equal
- School coverage group
  - School coverage: binary vector of units attempted
  - Grouping: clustered with k=3

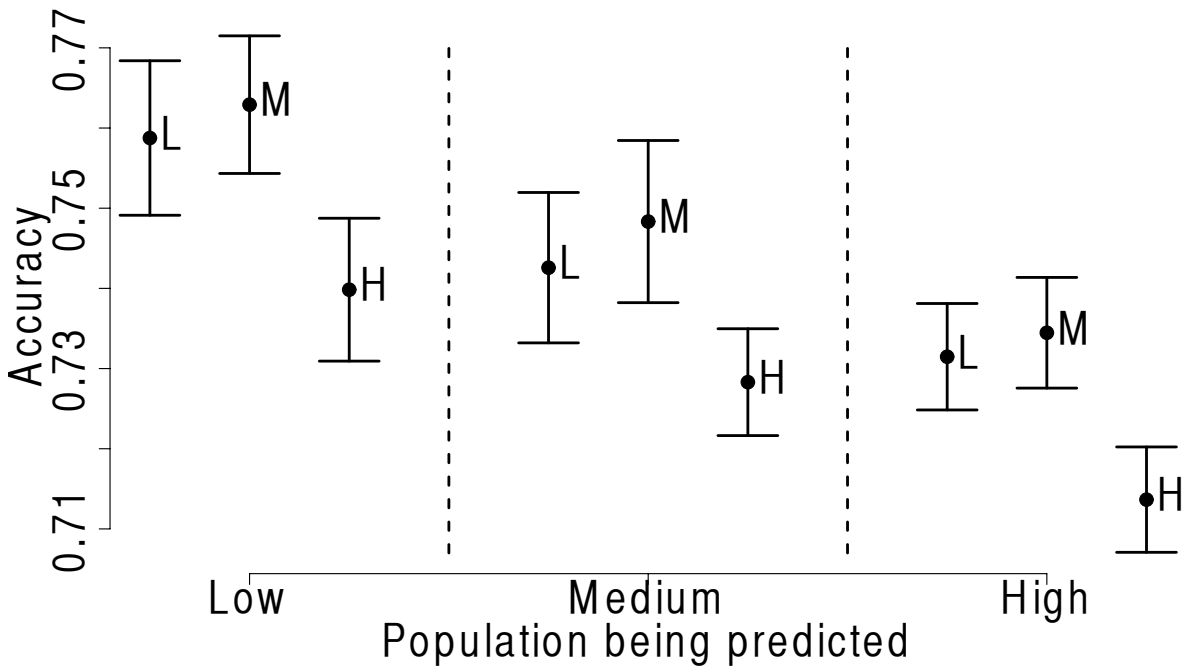
IS SCHOOL LOCALE AN EFFECTIVE GROUPING?

- Groups
  - Rural - 67
  - Suburban - 58
  - Urban - 107
- None of the group models have a *significant* advantage
- Models of schools in Rural group have a tendency to have a higher accuracy of prediction overall



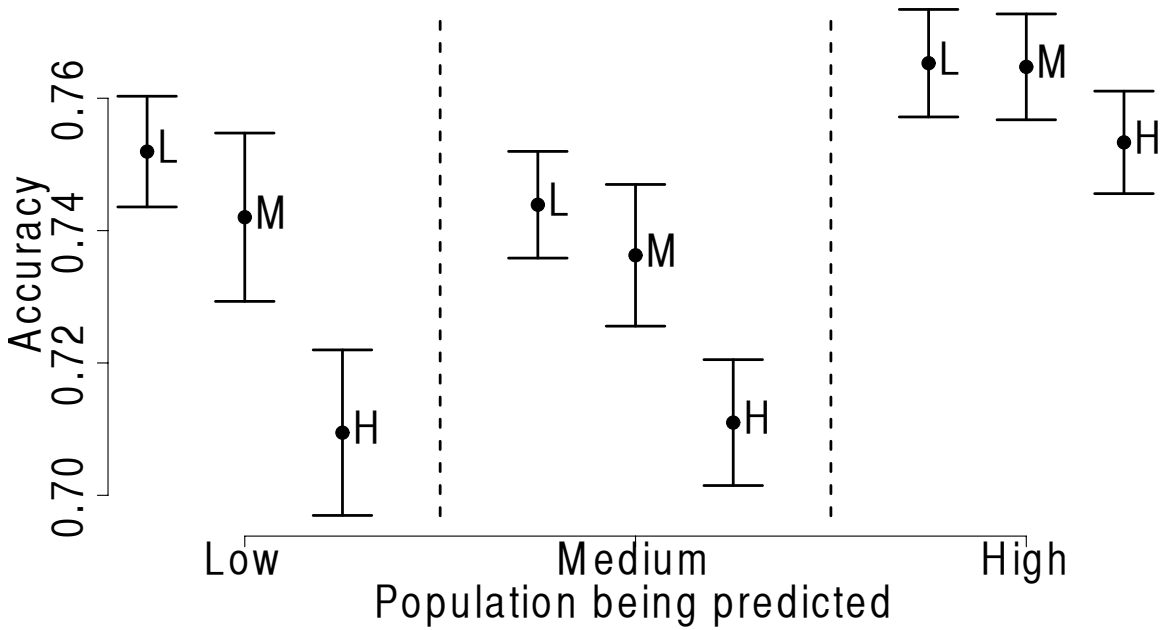
IS SCHOOL ENROLLMENT AN EFFECTIVE GROUPING?

- Group division
  - Low: <747students
  - Medium: [747, 1192)
  - High: ≥1192 students
- Model of high enrollment group of schools is *significantly* worse in 2/3 cases



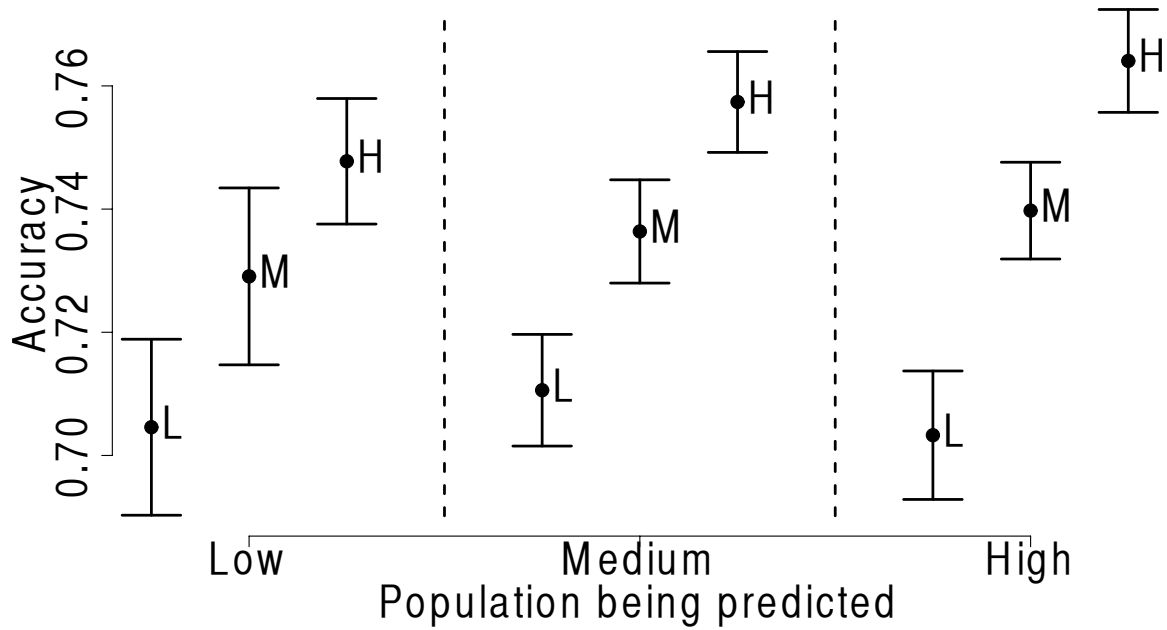
IS PERCENT STUDENTS ELIGIBLE FOR FREE AND RED. LUNCH AN EFFECTIVE GROUPING?

- Group division
  - Low: <48%
  - Medium: [48%, 70%)
  - High: ≥ 70%
- Model of high group of schools is *significantly* worse in 2/3 cases



IS AVERAGE STUDENTS UNITS ATTEMPTED AN EFFECTIVE GROUPING?

- Group division
  - Low: <5.8 units
  - Medium: [5.8, 9.2)
  - High: ≥ 9.2 units
- Model of schools with high number of units attempted is *significantly* better in 2/3 cases



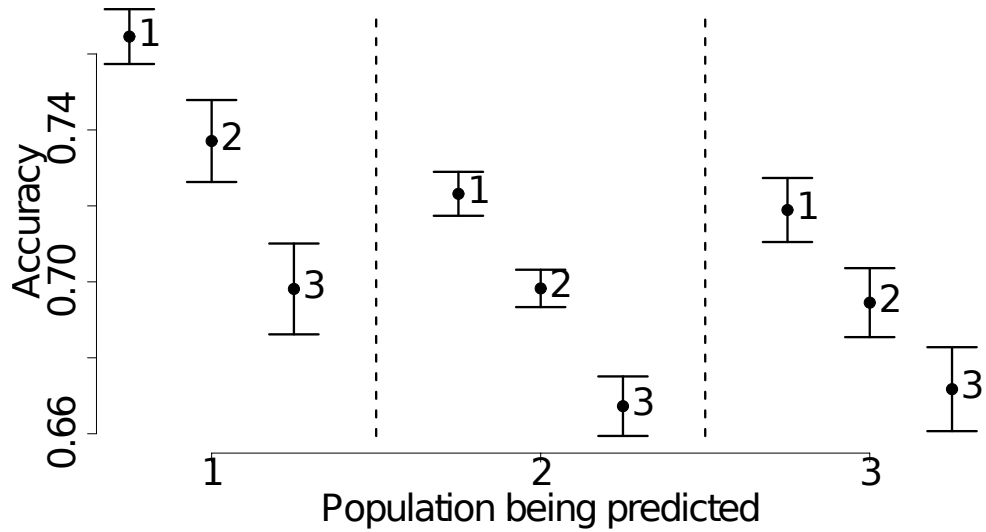
GROUPING FACTORS SUMMARY

Factor	Group	Result	Details
School locale	School a priori data	n/s	
%Free & Reduced Lunch		.	High is worse
Student-teacher ratio		n/s	
Enrollment		.	High is worse
Avg. student-units attempted	CL usage data	*	High is better
SE student-units attempted		n/s	
School coverage group		*	Cluster 2 is better
Avg. student intercept	Model values	*	High is better
Avg. skill intercept		.	Low is worse
Avg. skill slope		*	Medium is better

n/s – not significant, . – one group is sig worse, * - one group is sig better
(in all of the cases one group is better in 2/3 times)

IS THERE A *PROPERLY* EFFECTIVE GROUPING?

- The *good factors*
  - Avg. student units (CL)
  - Avg. student intercept (Model)
  - Avg. skill intercept (Model)
  - Avg. skill slope (Model)
- Continuous values of the *good factors* clustered to produce groups
  - Clustering using Ward's algorithm, as Yoav Bergner convinced us 2 days ago, is amazing



- All three groups are *cleanly* separated
  - Group 1: ~31000 students, ~120 schools
  - Group 2: ~13000 students, ~50 schools
  - Group 3: ~11000 students, ~65 schools

DISCUSSION. THE GOOD.

- We set out to discover distinct sub-groups
  - We found one group that effectively represents full sample
  - And there are multiple (partially overlapping) ways to define that group
- Result of *one group takes all* is conservative
  - When cross-predicting between school groups, missing skill values (unaddressed in the model) were replaced by a default value
  - This made inter-group differences less pronounced

DISCUSSION. THE BAD

- We cannot judge how representative is our dataset of the whole Carnegie Learning student population
  - Not all schools had logging enabled
  - Not all schools actually used the tutor
  - We only had NCES data for a subset of schools
- Let alone representativeness in the context of all K-12 schools in U.S. or the world

DISCUSSION. THE INTERESTING.

- Only good students should be considered for model-building?
  - It's not about student preparation, it's about student/teacher dedication
    - Recommended usage is 48hrs/semester
- Working hypothesis
  - Those who use the tutor more have an established track record (read, better suited for model building)
  - There is a few factors that influence the dedication
    - For example, Larger schools and lower SES could lead to worse student/teacher dedication to the Cognitive Tutor

Thank you!